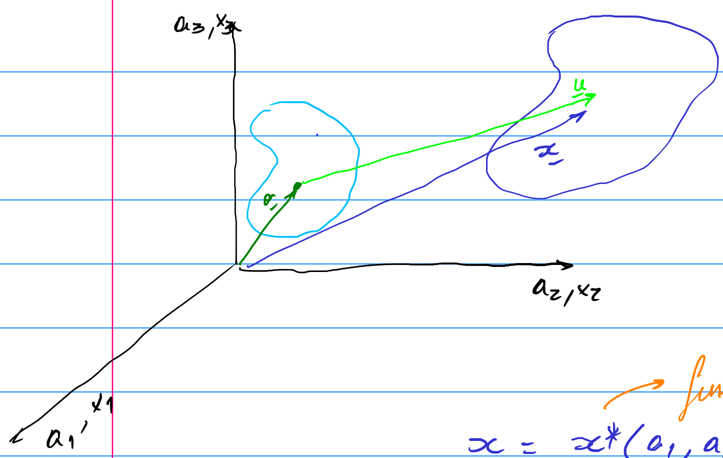
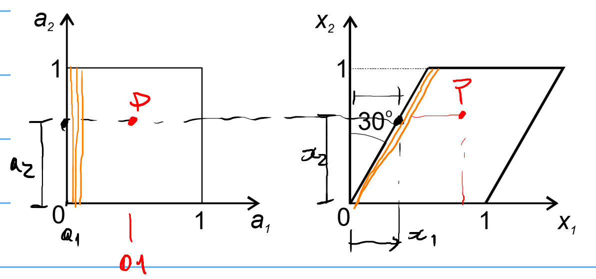


función vectorial que describe el movimiento

$$\underline{x} = \underline{x}^*(a_1, a_2, a_3, t)$$

$$\underline{a} = \underline{a}^*(x_1, x_2, x_3, t)$$

$$\underline{u} = \underline{x} - \underline{a}$$



$$x_2 = a_2$$

$$x_1 = 0 + tg\,30° \cdot a_2$$

$$x_2 = a_2$$

$$x_1 = a_1 + tg\,30°\, a_2$$

$$x_3 = a_3$$

$$E_{ij} = \frac{1}{2}\left(\delta_{\alpha\beta}\frac{\partial x_\alpha}{\partial a_i}\frac{\partial x_\beta}{\partial a_j} - \delta_{ij}\right), \quad = \frac{1}{2}\left(\frac{\partial x_\alpha}{\partial a_i}\cdot\frac{\partial x_\alpha}{\partial a_j} - \delta_{ij}\right)$$

$$e_{ij} = \frac{1}{2}\left(\delta_{ij} - \delta_{\alpha\beta}\frac{\partial a_\alpha}{\partial x_i}\frac{\partial a_\beta}{\partial x_j}\right), \quad = \frac{1}{2}\left(\delta_{ij} - \frac{\partial a_\alpha}{\partial x_i}\frac{\partial a_\alpha}{\partial x_j}\right)$$

son iguales

$$E = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}$$

$$E_{11} = \frac{1}{2}\left(\overbrace{\frac{\partial x_1}{\partial a_1}}^{1}\cdot\overbrace{\frac{\partial x_1}{\partial a_1}}^{1} + \overbrace{\frac{\partial x_2}{\partial a_1}}^{0}\cdot\overbrace{\frac{\partial x_2}{\partial a_1}}^{0} - \overbrace{\delta_{11}}^{1}\right) = 0$$

$$E_{12} = E_{21} = \frac{1}{2}\left(\overbrace{\frac{\partial x_1}{\partial a_1}}^{1}\cdot\overbrace{\frac{\partial x_1}{\partial a_2}}^{tg\,30°} + \overbrace{\frac{\partial x_2}{\partial a_1}}^{0}\cdot\overbrace{\frac{\partial x_2}{\partial a_2}}^{1} - \overbrace{\delta_{12}}^{0}\right) = \frac{1}{2}\,tg\,30°$$

$$E_{22} = \frac{1}{2}\left(\overbrace{\frac{\partial x_1}{\partial a_2}}^{tg\,30°}\cdot\overbrace{\frac{\partial x_1}{\partial a_2}}^{tg\,30°} + \overbrace{\frac{\partial x_2}{\partial a_2}}^{1}\cdot\overbrace{\frac{\partial x_2}{\partial a_2}}^{1} - \overbrace{\delta_{22}}^{1}\right) = \frac{1}{2}\,tg^2\,30°$$

$$E = \frac{1}{2}\begin{bmatrix} 0 & tg\,30° \\ tg\,30° & tg^2\,30° \end{bmatrix}$$

$x_2 = a_2$

$x_1 = a_1 + tg\,30^\circ a_2 = a_1 + tg\,30^\circ \cdot x_2$

$$\begin{cases} a_2 = x_2 \\ a_1 = x_1 - tg\,30^\circ x_2 \end{cases}$$